

3. 10. 1961

At 10.15 AM the first of the heavy rain fell
and continued for the next 2 hours.

At 12.15 PM the rain
ceased and the sun
came out.

The sun was out for about 1 hour
and the rain fell again at 1.15 PM.

The rain continued for
about 1 hour.

(The rain was very heavy)

At 2.15 PM the rain
ceased and the sun came out.

The sun was out for about 1 hour
and the rain fell again at 3.15 PM.

The rain continued for about 1 hour
and the sun came out at 4.15 PM.
The rain fell again at 5.15 PM.

Program: 66

Aim: write a C program with pattern of numbers

Resources Required: A PC with MS Turbo C++

Program:

// C program to print pattern with numbers

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i, j, n;
```

```
printf("enter the number: ");
```

```
scanf("%d", &n);
```

```
for(i=1; i<=n; i++)
```

```
{
```

```
for(j=1; j<=i; j++)
```

```
{
```

```
printf("%d", j);
```

```
}
```

```
printf("\n");
```

```
}
```

```
getch();
```

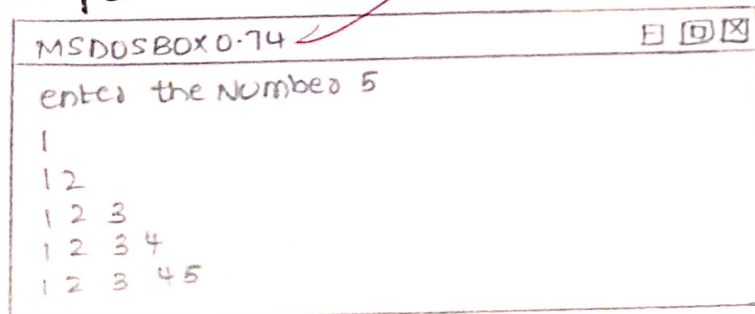
```
}
```

Save - click F2 to save it with .C extension.

Compile - click Alt+F9 to compile, rectify the errors if any

Execute - click ctrl+F9 to execute the program.

Output:



```
MSDOSBOX 0.74
enter the Number 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

The first step in the process of creating a new program is to determine the problem to be solved. This involves identifying the requirements of the program and the data that will be used. Once the problem is defined, the next step is to design a solution. This involves creating a flowchart or pseudocode that describes the logic of the program. The third step is to write the program code. This involves translating the design into a specific programming language. Finally, the program is tested and debugged to ensure it works correctly.

The process of creating a new program is a multi-step process that requires careful planning and execution. It is important to take the time to define the problem and design a solution before writing the code. This will help to ensure that the program is efficient and easy to maintain.

Program: 67

Aim: To write a C program to print Floyd's triangle.

Resources Required: A PC with MS Turbo C++

program:

// C program to print the Floyd's triangle.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main ()
```

```
{
```

```
int i, n, j, k=1;
```

```
clrscr();
```

```
printf("Enter numbers upto = ");
```

```
scanf("%d", &n);
```

```
for(i=1; i<=n; i++)
```

```
{
```

```
for(j=1; j<=i; j++)
```

```
{
```

```
printf("%d", k);
```

```
k++;
```

```
printf("\n");
```

```
getch();
```

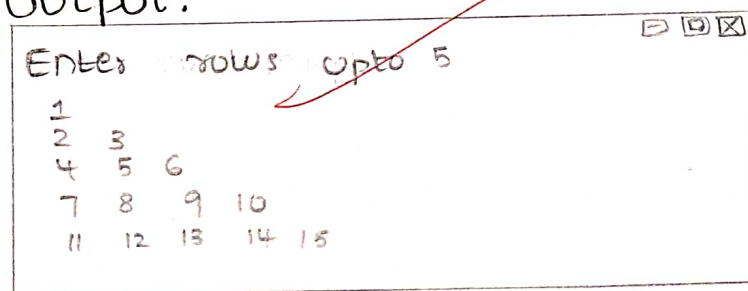
```
}
```

Save- click F2

Compile- click Alt+F9 to Compile, rectify the errors if any.

Execute- click Ctrl+F9 to execute the program.

Output:



```
Enter rows upto 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

1. The first step in the process of the scientific method is to ask a question.

2. The second step is to do background research on the topic.

3. The third step is to form a hypothesis, which is an educated guess about the answer to the question.

4. The fourth step is to design an experiment to test the hypothesis.

5. The fifth step is to conduct the experiment and collect data.

6. The sixth step is to analyze the data and draw a conclusion.

7. The seventh step is to communicate the results of the experiment to others.

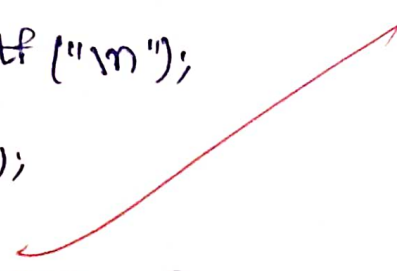
program: 68

Aim: write a C program to print pascal triangle

Resources Required: A PC with MS Turbo C++

program:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int rows, Coeff=1, space, i, j;
    clrscr();
    printf("enter no. of rows: ");
    scanf("%d", &rows);
    for(i=0; i<rows; i++)
    {
        for(space=1; space<=rows-i; space++)
            printf(" ");
        for(j=0; j<=i; j++)
        {
            Coeff = 1;
            else if if(j==0 || i==0)
                Coeff = Coeff * (i-j+1)/j;
            printf("%4d", Coeff);
        }
        printf("\n");
    }
    getch();
}
```

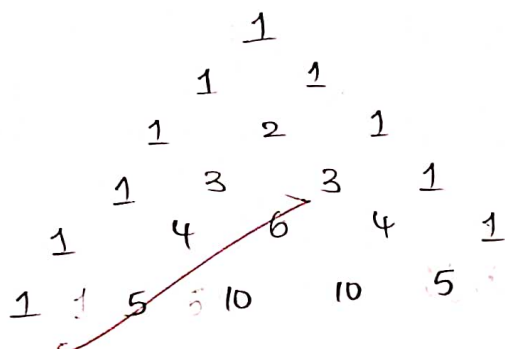


Save - click F2

Compile - click Alt+F9

Execute - click Ctrl+F9

enter no. of rows : 6



10 = 6 + 4

10 = 5 + 5

10 = 4 + 6

10 = 3 + 7

10 = 2 + 8

10 = 1 + 9

10 = 0 + 10

10 = -1 + 11

10 = -2 + 12

10 = -3 + 13

10 = -4 + 14

10 = -5 + 15

10 = -6 + 16

10 = -7 + 17

10 = -8 + 18

Program: 69

Aim: Write a C program to print pyramid of (*)

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int i, j, spc, rows, k;
    printf("\n enter no. of rows: ");
    scanf("%d", &rows);
    for(i=1; i<=rows; i++)
    {
        for(k=spc; k>=1; k--)
        {
            printf(" ");
        }
        for(j=1; j<=i; j++)
        {
            printf("*");
        }
        printf("\n");
        spc--;
    }
    getch();
}
```

Save -
Compile -
Execute -

enter no. of rows: 4

```
  *
 * *
* * *
* * * *
```


(a) The following are the main points to be noted:

1. The first point is that the

second point is that the

third point is that the

fourth point is that the

fifth point is that the

sixth point is that the

seventh point is that the

eighth point is that the

ninth point is that the

tenth point is that the

eleventh point is that the

twelfth point is that the

thirteenth point is that the

Program: 70

Aim: Write a C program to print Diamond pattern.

Resources Required: A PC with MS Turbo C++

Program:

// C program to print Diamond pattern.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i, j, rows;
```

```
int stars, spaces;
```

```
clrscr();
```

```
printf("Enter rows to print: ");
```

```
scanf("%d", &rows);
```

```
stars = 1
```

```
spaces = rows - 1;
```

```
/* Iterate through rows */
```

```
for(i=1; i<rows*2; i++)
```

```
{
```

```
/* print space */
```

```
for(j=1; j<spaces; j++)
```

```
printf(" ");
```

```
/* print Stars */
```

```
for(j=1; j<stars*2; j++)
```

```
printf(" ");
```

```
if(i<rows)
```

```
{
```

```
spaces --;
```

```
stars ++;
```

```
}
```

```
else
```

```
{ spaces ++;
```

```
stars --;
```

```
} getch(); }
```

Enter value of n : 3

Sum of series numbers = 0.500000.

Program: 75

Aim: Write a C program to print Sum of array Elements

program:

// C program to print Sum of array Elements.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int arr[100], i, sum=0;
```

```
int n;
```

```
clrscr();
```

```
printf("enter size of an array: ");
```

```
scanf("%d", &n);
```

```
printf("\n enter elements into array: ");
```

```
for(i=0; i<n; i++)
```

```
{ scanf("%d", &arr[i]);
```

```
}
```

```
for(i=0; i<n; i++)
```

```
{ sum = sum + arr[i];
```

```
}
```

```
printf("\n sum of array Elements = %d", sum);
```

```
getch();
```

```
}
```

Save -

Compile -

Execute -

enter size of an array:
enter elements into array: 21

63
77
67
62
35
37

sum of array elements =

Program: 76

Aim: write a C program for searching elements in array.

Resources Required: A PC with MS Turbo C++

program:

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int arr[20], i, j, temp, size;
```

```
clrscr();
```

```
printf("enter size of an array = ");
```

```
scanf("%d", &size)
```

```
printf("enter %d elements into array : \n", size);
```

```
for (i=0; i<size; i++)
```

```
{
```

```
scanf("%d", &arr[i]);
```

```
}
```

```
for (i=0; i<size; i++)
```

```
{ for (j=i+1; j<size; j++) {
```

```
if (arr[i] > arr[j])
```

```
{
```

```
temp = arr[i];
```

```
arr[i] = arr[j];
```

```
arr[j] = temp;
```

```
}
```

```
}
```

```
}
```

```
printf("\n array element in ascending order of array");
```

```
for (i=0; i<size; i++)
```

```
{
```

```
printf("%d", arr[i]);
```

```
}
```

```
getch();
```

```
}
```

Save - F2 Compile: Alt+F9 Execute: Ctrl+F9

Output:

```
MSDOS BOX 0.74  CPUSpeed: max 100%
enter the size of an array = 5
enter 5 elements in the array = 45
24
15
1
16
array elements in ascending order are
1
15
16
24
45
```

Program: 77

Aim: Write a C program to Search an array Element

Resources Required:

program:

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main ()
```

```
{
```

```
int arr[20], i, Search, Size;
```

```
clrscr();
```

```
printf("Enter size of an array");
```

```
scanf("%d", &size);
```

```
printf("\n Enter %d elements in array = ", size);
```

```
for(i=0; i<size; i++)
```

```
{  
scanf("%d", &arr[i]);
```

```
}
```

```
printf("\n Enter a number to Search = ");
```

```
scanf("%d", &Search);
```

```
for(i=0; i<size; i++)
```

```
{
```

```
if(arr[i] == Search)
```

```
{
```

```
printf("%d is present at location %d\n", Search, i+1);
```

```
break;
```

```
}
```

```
}
```

```
if(i == size)
```

```
printf("%d is not present in array\n", Search);
```

```
getch();
```

```
}
```

Save -

Compile -

Execute -

Output:

enter size of an array = 5

enter 5 elements in array = 12

23

34

45

56

enter a number to search = 34

34 is present at location 3

Program 78:

write a C program to find largest element in an array

Aim: To write a C program to find largest element in an array.

Resources Required: APC with MS Turbo C++

Program:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a[10], i;
    clrscr();
    printf("\n enter elements in array = ");
    a[0] = largest;
    for (i = 0; i < 5; i++)
    {
        scanf("%d", &a[i]);
    }
    for (i = 2; i < 5; i++)
    {
if (a[i] < a[i])
        if (a[0] < a[i])
        {
            a[0] = a[i];
        }
    }
    printf("\n largest element is %d", a[0]);
    getch();
}
```

Save - F2

Compile - Alt+F9

Execute - Ctrl+F9

1. The first part of the book is devoted to a description of the various types of...

of the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...

the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...
the various types of...

Program: 82

Aim: To write a C program to print Addition of Matrix.

Resources Required: A PC with Turbo C++

program:

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int a[2][2], b[2][2], c[2][2], i, j;
    clrscr();
    printf("enter elements into first matrix = ");
    for(i=0; i<2; i++)
    {
        for(j=0; j<2; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("\n enter elements into second matrix = ");
    for(i=0; i<2; i++)
    {
        for(j=0; j<2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }
    for(i=0; i<2; i++)
    {
        for(j=0; j<2; j++)
        {
            c[i][j] = 0;
        }
    }
    printf("\n Addition of two matrices = ");
    for(i=0; i<2; i++)
    {
        for(j=0; j<2; j++)
```

```

{
C[i][j] = a[i][j] + b[i][j];
}
}
for (i = 0; i < 2; i++)
{
for (j = 0; j < 2; j++)
{
printf("%3d", C[i][j]);
}
printf("\n");
}
getch();
}

```

Save - click F2 to save it with .c extension

Compile - click Alt+F9 to compile, rectify the errors if any.

Execute - click Ctrl+F9 to execute the program.

Output:

```

enter elements into first matrix:
1
2
3
4
enter elements into second matrix
5
6
7
8
Addition of matrix elements
6      8
10     12

```

1. (11/11/11) 1. (11/11/11)

2. (11/11/11) 2. (11/11/11)

3. (11/11/11) 3. (11/11/11)

4. (11/11/11) 4. (11/11/11)

5. (11/11/11) 5. (11/11/11)

6. (11/11/11) 6. (11/11/11)

7. (11/11/11) 7. (11/11/11)
8. (11/11/11) 8. (11/11/11)
9. (11/11/11) 9. (11/11/11)
10. (11/11/11) 10. (11/11/11)
11. (11/11/11) 11. (11/11/11)
12. (11/11/11) 12. (11/11/11)
13. (11/11/11) 13. (11/11/11)
14. (11/11/11) 14. (11/11/11)
15. (11/11/11) 15. (11/11/11)
16. (11/11/11) 16. (11/11/11)
17. (11/11/11) 17. (11/11/11)
18. (11/11/11) 18. (11/11/11)
19. (11/11/11) 19. (11/11/11)
20. (11/11/11) 20. (11/11/11)

Program: 83

Aim: To write a C program to print Subtraction of a Matrices.

Resources Required:

program:

// C program to print Subtraction of Matrix elements.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int a[2][2], b[2][2], c[2][2], i, j;
```

```
clrscr();
```

```
printf("enter elements into first matrix = ");
```

```
for(i=0; i<2; i++)
```

```
{
```

```
for(j=0; j<2; j++)
```

```
{
```

```
scanf("%d", a[i][j]);
```

```
}
```

```
}
```

```
printf("enter elements in second matrix = ");
```

```
for(i=0; i<2; i++)
```

```
{
```

```
for(j=0; j<2; j++)
```

```
{
```

```
scanf("%d", b[i][j]);
```

```
}
```

```
}
```

```
for(i=0; i<2; i++)
```

```
{
```

```
for(j=0; j<2; j++)
```

```
{
```

```
c[i][j] = 0;
```

```
}
```

```
}
```

```
printf("\n subtraction of two matrices = ");
```



```

for(i=0; i<2; i++) {
    for(j=0; j<2; j++)
    {
        c[i][j] = a[i][j] - b[i][j];
    }
}
for(i=0; i<2; i++)
{
    for(j=0; j<2; j++)
    {
        printf("%3d", c[i][j]);
    }
    printf("\n");
}
getch();
}

```

Save: Click F2 to save it with .c extension.

Compile: Click Alt+F9 to compile, rectify the errors if any

Execute: Click Ctrl+F9 to execute the program.

Output:

```

MS-DOS BOX 0.74  CPU speed: max 100%
enter elements into first elements
5
6
7
8
enter elements into second matrix
2
2
2
2
Subtraction of matrix elements
3 4
5 6

```

DATE: 10/10/13
PAGE: 10

NAME: [illegible]

TOPIC: [illegible]

DATE: 10/10/13

NAME: [illegible]

DATE: 10/10/13

NAME: [illegible]

1. The first step in the process of creating a new product is to identify a need or want in the market. This is often done through market research, which can involve surveys, focus groups, and other methods of gathering information from potential customers. Once a need or want has been identified, the next step is to develop a concept for a product that will satisfy that need or want. This involves brainstorming ideas and selecting the most promising one. The final step in the process is to create a prototype of the product, which can be used to test the concept and make any necessary adjustments before moving forward with production.

Program: 84

Aim: To write a C program to print product of Matrix.

Re
program:

// C program to print the multiplication of 2 Matrices

```
#include <stdio.h>
```

```
#include <conio.h> #define max 10
```

```
void main()
```

```
{  
    int a[max][max], b[max][max], i, j, k, row, col;
```

```
    clrscr();
```

```
    printf("row matrix is:");
```

```
    scanf("%d", &row);
```

```
    printf("column matrix is:");
```

```
    scanf("%d", &col);
```

```
    for(i=0; i<row; i++)
```

```
    {  
        for(j=0; j<col; j++)
```

```
        {  
            scanf("%d", &a[i][j]);
```

```
            printf("enter b matrix");
```

```
            for(i=0; i<row; i++)
```

```
            {  
                for(j=0; j<col; j++)
```

```
                {  
                    scanf("%d", &b[i][j]);
```

```
                }
```

```
            }
```

```
        for(i=0; i<row; i++)
```

```
        {  
            for(j=0; j<col; j++)
```

```
            {  
                c[i][j] = 0;
```

```
            }
```

output:

enter elements into first matrix = 1

2

3

4

enter elements into second matrix = 5

6

7

8

multiplication of matrix...

10 24

42 64


```
printf("In multiplication of two matrix... \n");
```

```
for (i=0; i<2; i++)
```

```
{  
  for (j=0; j<2; j++)
```

```
{  
  for (k=0; k<2; k++)
```

```
{  
  c[i][j] = c[i][j] + a[i][k] * b[k][j];
```

```
  a[i][k] * b[k][j];  
  b[k][j];
```

```
}  
}  
for (i=0; i<2; i++)
```

```
{  
  for (j=0; j<2; j++)
```

```
{  
  printf("%3d", c[i][j]);
```

```
}  
  printf("\n");
```

```
}  
getch();
```

Save - click F2

Compile - click ALT+F9

Execute - click CTRL+F9

For the multiplication of two arrays (1D arrays)

for (i = 0; i < n; i++)

for (k = 0; k < n; k++)

for (j = 0; j < n; j++)

arr[i][j] = arr[i][k] + arr[k][j];

for (i = 0; i < n; i++)

for (j = 0; j < n; j++)

arr[i][j] = arr[i][j] + arr[i][k];

arr[i][j] = arr[i][j] + arr[k][j];

arr[i][j] = arr[i][j] + arr[k][j];

arr[i][j] = arr[i][j] + arr[k][j];

arr[i][j] = arr[i][j] + arr[k][j];

Program: 85

Write a C program to find transpose of a matrix.

Aim: Write a C program to find Transpose of a matrix.

Resources Required: A PC with Turbo C Application.

program:

```
#include <stdio.h>
#include <conio.h>
Void main()
{
    int a[3][3], b[3][3], i, j;
    clrscr();
    printf("Enter value for matrix:");
    for(i=0; i<3; i++)
    {
        for(j=0; j<3; j++) {
            scanf("%d", &a[i][j]);
        }
        printf("Matrix: \n");
        for(i=0; i<3; i++)
        {
            for(j=0; j<3; j++) {
                printf("%d", a[i][j]);
            }
            printf("\n");
        }
        for(i=0; i<3; i++)
        {
            for(j=0; j<3; j++)
                b[i][j] = a[j][i];
        }
        printf("Transpose Matrix: \n");
        for(i=0; i<3; i++)
        {
```

```

{
for(j=0; j<3; j++)
printf(" %d", b[i][j]);
printf("\n");
}
getch();
}

```

Save - Click F2

Compile - ALT+F9

Execute - CTRL+F9

Output

MSDOS BOX 0.74

Enter values for matrix

1

2

3

4

5

6

9 Transpose of matrix

1 4
2 5
3 6